

Multiple engine plotting Penguins

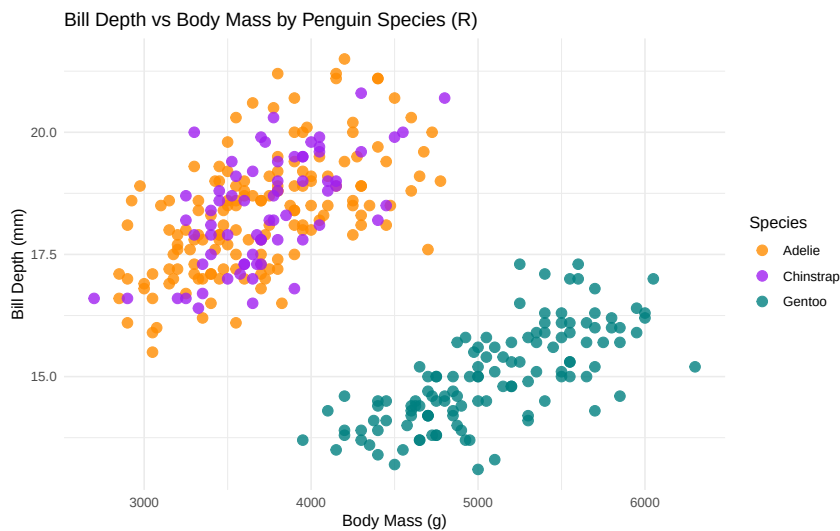
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0.1 R Visualization

```
# Create scatter plot with R
ggplot(penguins_clean, aes(x = body_mass_g, y = bill_depth_mm, color = species)) +
  geom_point(alpha = 0.8, size = 3) +
  scale_color_manual(values = c("Adelie" = "darkorange", "Chinstrap" = "purple", "Gentoo" = "teal")) +
  labs(
    title = "Bill Depth vs Body Mass by Penguin Species (R)",
    x = "Body Mass (g)",
    y = "Bill Depth (mm)",
    color = "Species"
  ) +
  theme_minimal()
```



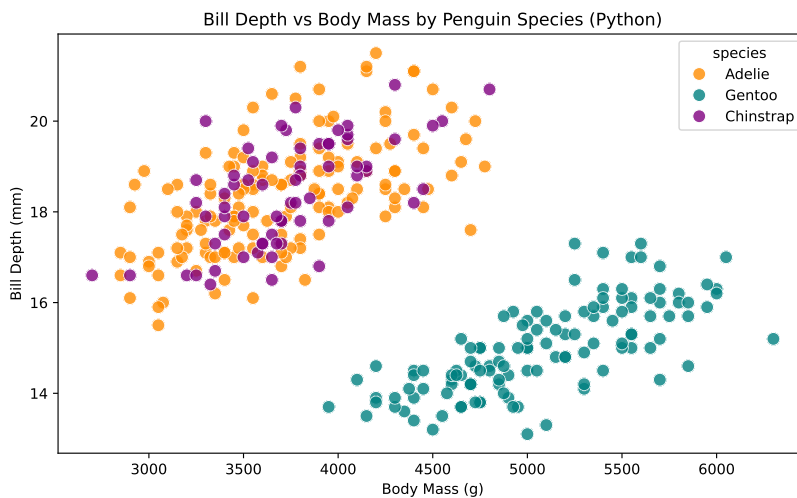
0.2 Python Visualization

```
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Access the penguins data from R
penguins = r.py_penguins

# Create scatter plot with Python
plt.figure(figsize=(8, 5))
sns.scatterplot(
    data=penguins,
    x="body_mass_g",
    y="bill_depth_mm",
    hue="species",
    palette={"Adelie": "darkorange", "Chinstrap": "purple", "Gentoo": "#008080"},
    alpha=0.8,
    s=80
)

plt.title("Bill Depth vs Body Mass by Penguin Species (Python)")
plt.xlabel("Body Mass (g)")
plt.ylabel("Bill Depth (mm)")
plt.tight_layout()
plt.show()
```



0.3 Julia Visualization

```
try
    # Basic Julia plot - only runs if Julia is available
    using CSV
    using DataFrames
    using Plots
```



```

# Force the GR backend and set output format
ENV["GKSwstype"] = "100" # Ensure headless plotting works
Plots.gr() # Explicitly use GR backend

# Check if file exists
if isfile("penguins_clean.csv")
    # Read data and create scatter plot
    penguins = CSV.read("penguins_clean.csv", DataFrame)

    # Convert species to string type
    penguins.species = string.(penguins.species)

    # Create scatter plot with hex color codes and store in variable
    plt = scatter(
        penguins.body_mass_g,
        penguins.bill_depth_mm,
        group=penguins.species,
        markersize=6,
        alpha=0.8,
        markerstrokewidth=0,
        color=["#FFA500", "#800080", "#008080"], # hex codes for orange, purple, teal
        xlabel="Body Mass (g)",
        ylabel="Bill Depth (mm)",
        title="Bill Depth vs Body Mass by Penguin Species (Julia)",
        legend=:bottomright,
        size=(800, 500),
        dpi=300,
        fmt=:png # Force PNG output format
    )

    # Explicitly display the plot
    display(plt)

    # Return the plot as the last expression
    plt
else
    println("Error: penguins_clean.csv file not found")
    # Create a simple error plot
    plot(title="Error: Data file not found", legend=false)
end
catch e
    println("Julia error: ", e)
    # Return something that can be displayed
    plot(title="Julia Error Occurred", legend=false)
end

```

Plot{Plots.GRBackend() n=3}

